

VLAN Configuration & Segmentation

Overview

VLANs were implemented throughout the enterprise network to logically separate different categories of devices and traffic.

The network uses VLANs for PCs, phones, servers, wireless clients, and network management. Layer 3 switching and inter-VLAN routing allow communication between approved VLANs while maintaining logical segmentation.

VLAN Inventory

VLAN ID	Name	Purpose	Network
10	PCs	User/client devices	10.1.0.0/24 or 10.3.0.0/24
20	Phones	Voice devices	10.2.0.0/24 or 10.4.0.0/24
30	Servers	Server infrastructure	10.5.0.0/24
40	Wi-Fi	Wireless clients	10.6.0.0/24
99	Management	Network management	10.0.0.0/28 or 10.0.0.16/28

A-Side VLAN Configuration

DSW-A1

DSW-A1 has the following active VLANs:

VLAN	Name	Status
10	PCs	Active
20	Phones	Active
40	Wi-Fi	Active
99	Management	Active

The `show vlan brief` output confirms that VLANs 10, 20, 40, and 99 are configured and active on DSW-A1.

VLAN Functions

VLAN 10 — PCs

Used to separate PC/user traffic from other device categories.

VLAN 20 — Phones

Used to logically separate voice traffic from general user traffic.

VLAN 40 — Wi-Fi

Used for wireless client traffic on the A-side network.

VLAN 99 — Management

Used for network management traffic and switch management interfaces.

DSW-A2

DSW-A2 uses the same VLAN structure as DSW-A1:

VLAN	Name	Status
10	PCs	Active
20	Phones	Active
40	Wi-Fi	Active
99	Management	Active

The configuration output confirms VLANs 10, 20, 40, and 99 as active.

The corresponding Layer 3 interfaces provide connectivity for the A-side VLAN networks, including 10.1.0.0/24, 10.2.0.0/24, and 10.6.0.0/24.

B-Side VLAN Configuration

DSW-B1

DSW-B1 uses the following active VLANs:

VLAN	Name	Status
10	PCs	Active
20	Phones	Active
30	Servers	Active
99	Management	Active

The `show vlan brief` output confirms VLANs 10, 20, 30, and 99 as active on the B-side network.

VLAN Functions

VLAN 10 — PCs

Used for PC/user devices on the B-side network.

VLAN 20 — Phones

Used to separate voice devices and traffic.

VLAN 30 — Servers

Used to isolate server infrastructure from user and voice networks.

VLAN 99 — Management

Used for network management and infrastructure administration.

DSW-B2

DSW-B2 follows the same VLAN structure as DSW-B1:

VLAN	Name	Status
10	PCs	Active
20	Phones	Active
30	Servers	Active
99	Management	Active

The configuration output confirms VLANs 10, 20, 30, and 99 as active.

The routing information also confirms that VLAN 10, VLAN 20, and VLAN 30 correspond to the directly connected 10.3.0.0/24, 10.4.0.0/24, and 10.5.0.0/24 networks.

VLAN-to-Network Mapping

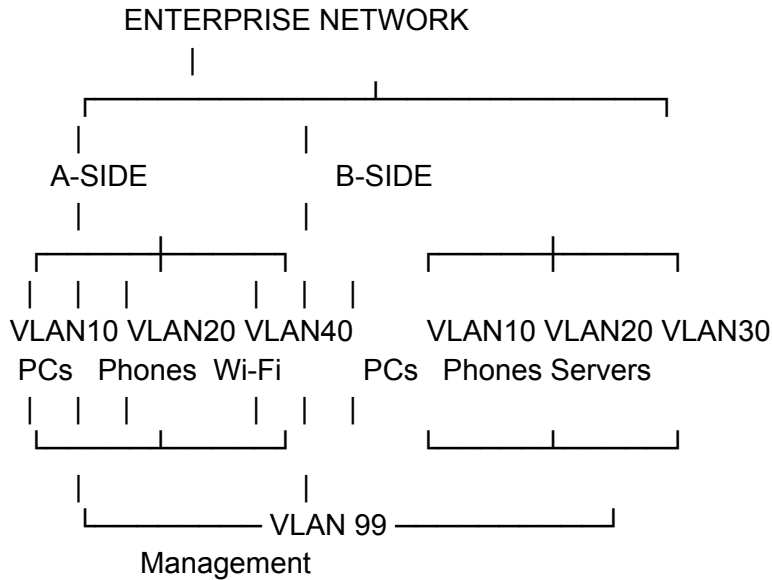
The enterprise topology uses different address ranges on the A-side and B-side while maintaining consistent VLAN purposes.

Network Area	VLAN	Purpose	IPv4 Network
A-Side	10	PCs	10.1.0.0/24
A-Side	20	Phones	10.2.0.0/24
A-Side	40	Wi-Fi	10.6.0.0/24
A-Side	99	Management	10.0.0.0/28
B-Side	10	PCs	10.3.0.0/24
B-Side	20	Phones	10.4.0.0/24
B-Side	30	Servers	10.5.0.0/24
B-Side	99	Management	10.0.0.16/28

The routing tables confirm the A-side VLAN networks as directly connected on VLAN 10, VLAN 20, and VLAN 40, while the B-side uses VLAN 10, VLAN 20, and VLAN 30 for its local networks.

VLAN Segmentation Design

The VLAN structure separates network traffic based on device and operational purpose.



This segmentation provides logical separation between user, voice, wireless, server, and management traffic.

Inter-VLAN Routing

Layer 3 switching is used to provide routing between VLANs.

The distribution switches use Switched Virtual Interfaces (SVIs) as Layer 3 gateways for the VLAN networks.

For example, the A-side distribution switches have directly connected networks including:

10.1.0.0/24 → VLAN 10
10.2.0.0/24 → VLAN 20
10.6.0.0/24 → VLAN 40

The B-side distribution switches have:

10.3.0.0/24 → VLAN 10
10.4.0.0/24 → VLAN 20
10.5.0.0/24 → VLAN 30

The routing tables confirm these VLAN-to-network relationships.

VLAN Verification

The following Cisco IOS command was used to verify VLAN configuration:

```
show vlan brief
```

This command verifies:

- VLAN ID
- VLAN name
- VLAN status
- Access-port assignments

Example verification:

```
DSW-A1#show vlan brief
```

Expected configured VLANs include:

```
10 PCs      active
20 Phones   active
40 Wi-Fi    active
99 Management active
```

On the B-side:

```
DSW-B2#show vlan brief
```

Expected configured VLANs include:

```
10 PCs      active
20 Phones   active
30 Servers   active
99 Management active
```

Security Considerations

VLAN segmentation reduces the amount of unnecessary Layer 2 traffic between different device groups and provides a foundation for implementing network security controls.

The separation of:

- User devices
- Voice devices
- Servers
- Wireless clients
- Management traffic

allows security policies such as ACLs and inter-VLAN access controls to be applied according to the role of each network.

VLAN segmentation should not be treated as a complete security boundary by itself. Additional controls such as ACLs, secure management protocols, and appropriate switch security features are used to control access between network segments.

Summary

The lab implements VLAN segmentation across an enterprise-style network using VLANs 10, 20, 30, 40, and 99.

The A-side network uses VLAN 40 for Wi-Fi, while the B-side network uses VLAN 30 for server infrastructure. VLAN 99 provides management segmentation across the environment.

Layer 3 SVIs provide gateway functionality for the VLAN networks, while OSPF provides dynamic routing between the distribution and core infrastructure.

This design demonstrates practical experience with:

- VLAN creation
- VLAN segmentation
- Layer 3 switching
- SVI configuration
- Inter-VLAN routing
- Network management segmentation
- Cisco IOS verification
- Enterprise network design